MLA 0204 - FOML

CIA - 1

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(1) Given $P(C) = 0.03$

$P(+|C) = 0.90$

$P(-|\neg C) = 0.94$

Loss of not treating diseased patient = 50

Loss of treating healthy patient = 2

correct decision cost = 0

Calculate Prior Probabilities and Likelihoods

$P(\neg C) = 1 - 0.03 = 0.97$

$P(-|C) = 1 - 0.90 = 0.10$

$P(+|\neg C) = 1 - 0.94 = 0.06$

Calculate Evidence

$P(+) = P(+|C)\,P(C) + P(+|\neg C)\,P(\neg C)$

$P(+) = 0.90\,(0.03) + 0.06\,(0.97) = 0.0852$

$P(-) = 1 - 0.0852 = 0.9148$

Calculate Posterior Probabilities

$P(C|+) = \frac{0.09 \times 0.03}{0.0852} = 0.3169$

$P(\neg C|+) = 1 - 0.3169 = 0.6831$

$$P(C|-) = \frac{0.10 \times 0.03}{0.9148} = 0.0033$$

$$P(\neg C|-) = 1 - 0.0033 = 0.9967$$

Calculate conditional Risky

$$R(\text{Treat}|+) = 2 \times 0.6831 = 1.3662$$

$$R(\text{No Treat}|+) = 50 \times 0.3169 = 15.845$$

$$R(\text{Treat}|-) = 2 \times 0.9967 = 1.9934$$

$$R(\text{No Treat}|-) = 50 \times 0.0033 = 0.1640$$

minimum Risk decisions

positive result → Treat (1.3662 is minimum)

negative result → Do not Treat (0.1640 is minimum)

minimum error (0-1 Loss) Rule

Since $P(C|+) = 0.3169 < 0.5$, patient is classified as healthy

Expected loss ⇒ $R = 50 \times 0.03 = \boxed{1.5}$

(1)

Given $P(X) = (0.32, 0.38, 0.30)$

$P(Y) = (0.68, 0.32)$

$Q = (0.5, 0.3, 0.2)$

Calculate marginal Probability

$P(X) = (0.32, 0.38, 0.30)$

$P(Y) = (0.68, 0.32)$

Calculate Entropy $H(X)$ and $H(Y)$

$$H(X) = -\sum P(x) \log_2 P(x)$$

$H(X) = 0.5260 + 0.5304 + 0.5211 = 1.5776$

$H(Y) = 0.3784 + 0.5260 = 0.9044$

Calculate Joint Entropy

$$H(X, Y) = \sum P(x, y) \log_2 P(x, y)$$

$= 0.5211 + 0.1129 + 0.5142 + 0.3322 + 0.3322 + 0.4644$

$= 2.2770$

Calculate KL Divergence

$$D_{KL}(P_X || Q) = H(P_X, Q) - H(X)$$

$= 1.6766 - 1.5776 = 0.0990$

Calculate conditional fault probability

$P(Y = yes \mid X = High) = 0.6667$

$P(Y = yes \mid X = Low) = 0.0625$

$P(Y = yes \mid X = medium) = 0.2632$

(3)

i) Given $X = \{48, 52, 55, 47, 53, 51\}$, $n = 6$

ML mean $\mu = \frac{\Sigma x}{n} = \frac{306}{6} = 51$

ML variance $\sigma^2 = \frac{\Sigma (x-\mu)^2}{n} = \frac{46}{6} = 7.667$

unbiased variance $s^2 = \frac{46}{6-1} = 9.20$

Calculate $P(x > 55)$

$z = \frac{55 - 51}{2.769} = 1.445$

$P(x > 55) = 1 - \phi(1.445) = 0.07432$

Calculate $P(48 < x < 54)$

$$z = \frac{54 - 51}{2.769} = 1.083$$

$$P = 2\phi(1.083) - 1 = \boxed{0.7214}$$

ii) Given

$$(x, y) = (10, 25), (20, 38), (30, 52), (40, 61), (50, 79)$$

$$\lambda = 500$$

Mean

$$\bar{x} = 30, \quad \bar{y} = 51$$

$$S_{xx} = 1000, \quad S_{xy} = 1310$$

Regression coefficients

$$w_1 = \frac{1310}{1000} = 1.31$$

$$w_0 = 51 - (1.31 \times 30) = 11.7$$

$$\boxed{\bar{y} = 11.7 + 1.31x}$$

SSE

$$SSE = 0.04 + 0.01 + 1 + 9.61 + 3.24$$

$$SSE = 13.90$$

R^2

$$R^2 = 1 - \frac{13.90}{1730} = 0.9920$$

Predict at $x = 60$

$$\bar{y} = 11.7 + 1.31\,(60) = 90.3 \text{ ms}$$

Ridge Regression

$$w_1 = \frac{1310}{1000 + 500} = 0.8733$$

$$w_0 = 51 - (0.8733 \times 30) = 24.8$$

Ridge model

$$\bar{y} = 24.8 + 0.8733x$$

$$\bar{y}(60) = 24.8 + 0.8733\,(60) = 77.2 \text{ ms}$$

Conclusion

Ordinary Regression predicts 90.3 ms while Ridge Regression predicts 77.2 ms. Ridge Regression Reduces Slope but causes under-fitting due to high regularization

1) Given $N_{Spam} = 40$ $N_{legit} = 60$

offer = (28, 6)

Free = (24, 3)

meeting = (4, 33)

Calculate prior probabilities

$$P(Spam) = \frac{40}{100} = 0.40$$

$$P(legit) = \frac{60}{100} = 0.60$$

Smooth probability for Spam

$$P(offer \mid Spam) = \frac{29}{42} = 0.6905$$

$$P(free \mid Spam) = \frac{25}{42} = 0.5952$$

$$P(meeting \mid Spam) = \frac{5}{42} = 0.1190$$

Smooth probability for Legitimate

$$P(offer \mid legit) = \frac{7}{62} = 0.1129$$

$$P(free \mid legit) = \frac{4}{62} = 0.0645$$

$$P(\text{meeting} \mid \text{legit}) = \frac{34}{62} = 0.5484$$

Calculate Scores

$$\text{Score (Spam)} = 0.40\,(0.6905)(0.5952)\,[1 - 0.1190]$$

$$= 0.1448$$

$$\text{Score (legit)} = 0.60\,(0.1129)(0.0645)(1 - 0.5484)$$

$$= 0.00197$$

Cost Sensitive classification

$$\text{Threshold} = \frac{5}{5+1} = 0.8333$$

Since $0.9866 > 0.8333$, email is Spam

Performance metrics

TP = 45, FN = 15, FP = 10, TN = 130

$$\text{Accuracy} = \frac{45 + 130}{200} \times 100 = 87.5\,\%$$

Precision $= \frac{45}{55} \times 100 = 81.82\ \%$

Recall $= \frac{45}{60} \times 100 = 75\ \%$

F1 $= \frac{90}{115} \times 100 = 78.26\ \%$

Misclassification Cost

Cost $= 5(FP) + 1(FN) = 5(10) + 15 = 65$

10

FPR $= \frac{10}{140} \times 100 = 7.14\ \%$

(5) Given $x = (0.5, 1.0)$, $t = 1$, $\eta = 0.5$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Hidden Layer outputs

$z_1 = 0.4(0.5) - (0.2)(1) + 0.1 = 0.100$

$z_2 = 0.3(0.5) + (0.5)(1) - 0.1 = 0.550$

Hidden Activations

$h_1 = \sigma(0.100) = 0.5250$

$$h_2 = \sigma(0.550) = 0.6341$$

Network output

$$z_0 = 0.6(0.5250) - 0.4(0.6341) + 0.2$$

$$y = \sigma(0.2613) = \boxed{0.5650}$$

Error

$$E = \frac{1}{2}(t-y)^2$$

$$E = \frac{1}{2}(1-0.5650)^2 = 0.0946$$

Delta

$$\delta_0 = (y-t)\,y(1-y)$$

$$= (-0.4350)(0.5650)(0.4350)$$

$$= -0.1069$$

Hidden Layer deltas

$$\delta_1 = \delta_0 v_1 h_1 (1-h_1) = -0.0160$$

$$\delta_2 = \delta_0 v_2 h_2 (1-h_2) = 0.0099$$

update output Layer weights

$$v_1 = 0.6281 \qquad v_2 = -0.3661 \qquad b_3 = 0.2535$$

Calculate new output and Error

After updating weights

$$y_{new} = 0.5878$$

$$E_{new} = \frac{1}{2}(1 - 0.5878)^2$$

$$E_{new} = 0.0850$$

Conclusion

After one back propagation update, the network output increases from 0.5650 to 0.5878 and error decreases from 0.0946 to 0.0850

10